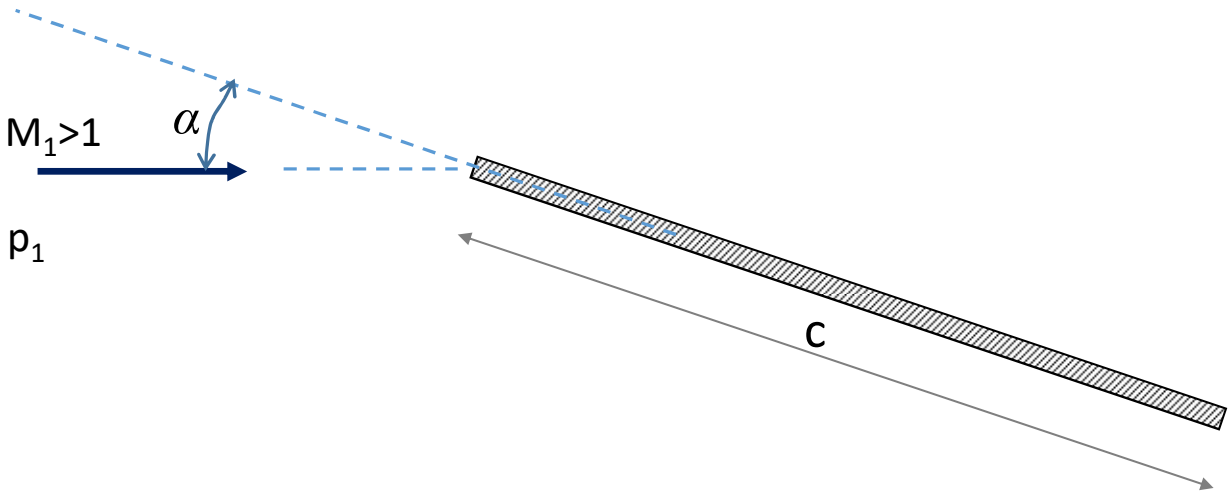
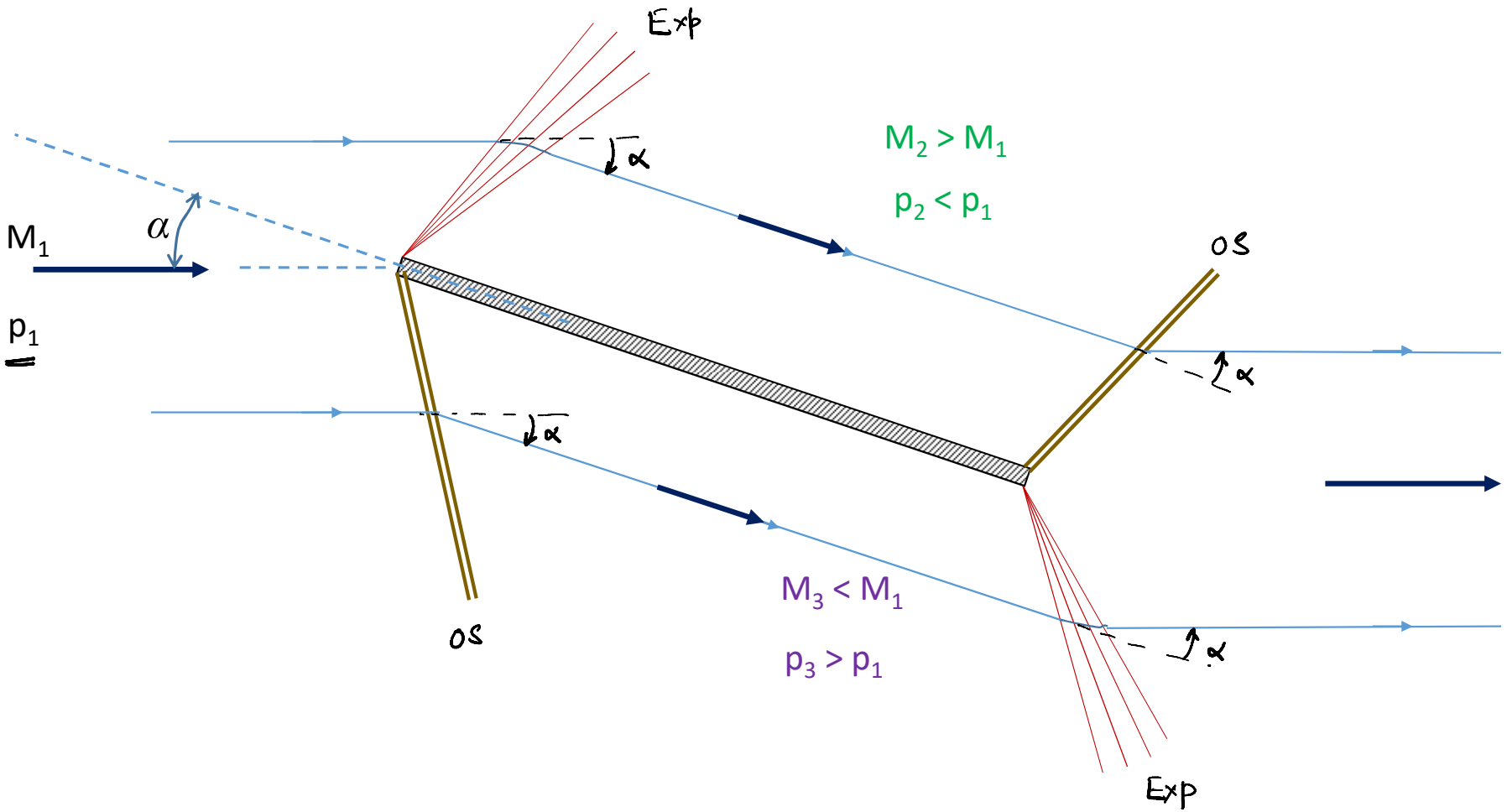
SHOCK-EXPANSION THEORY: APPLICATIONS TO SUPERSONIC AIRFOILS

Consider a thin flat plate of length c kept at an angle of attack α in a supersonic flow

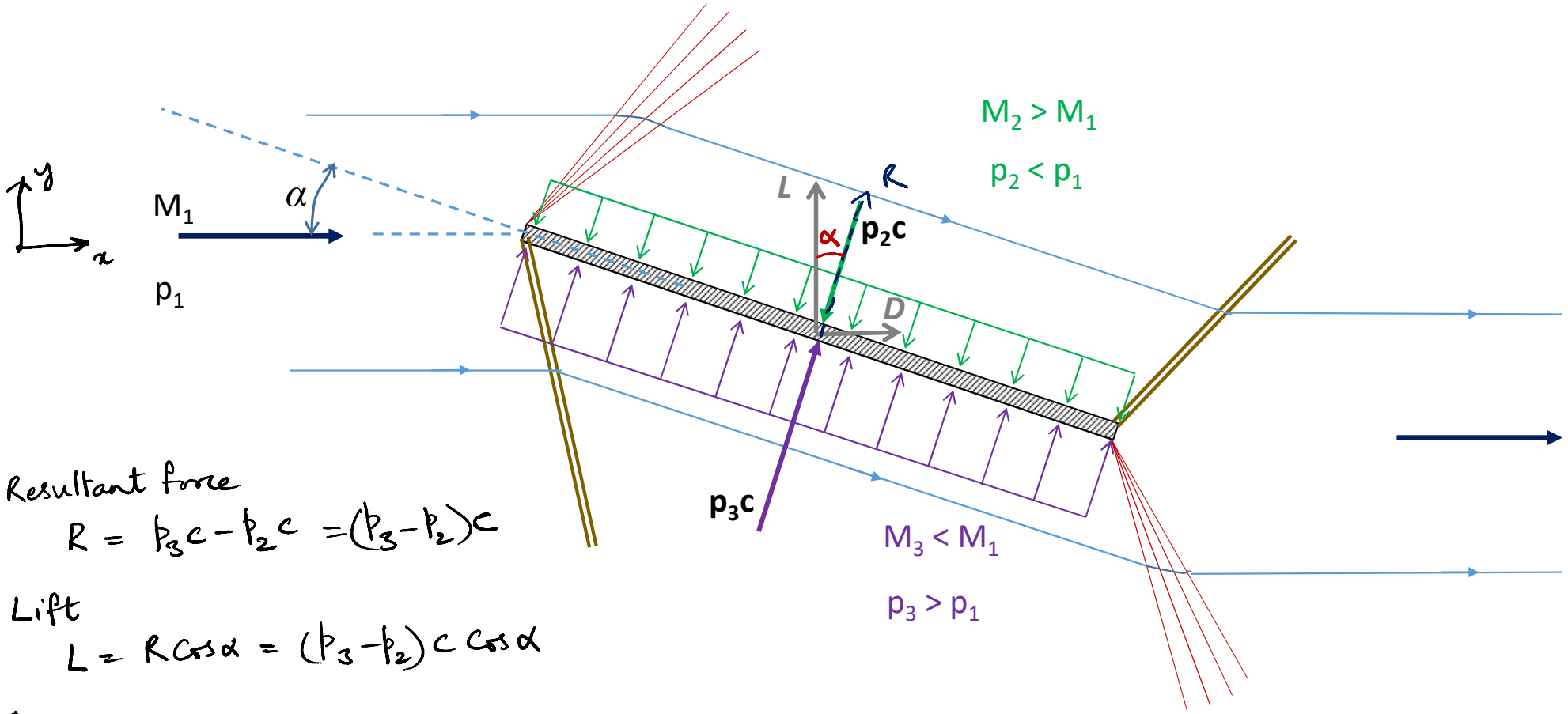


Consider a thin flat plate kept at an angle of attack in a supersonic flow



Consider a thin flat plate kept at an angle of attack in a supersonic flow

Length of the plate = c



Resultant force

$$R = p_3 c - p_2 c = (p_3 - p_2) c$$

Lift

$$L = R\cos\alpha = (p_3 - p_2) c \cos\alpha$$

Drag

$$D = R\sin\alpha = (p_3 - p_2) c \sin\alpha$$

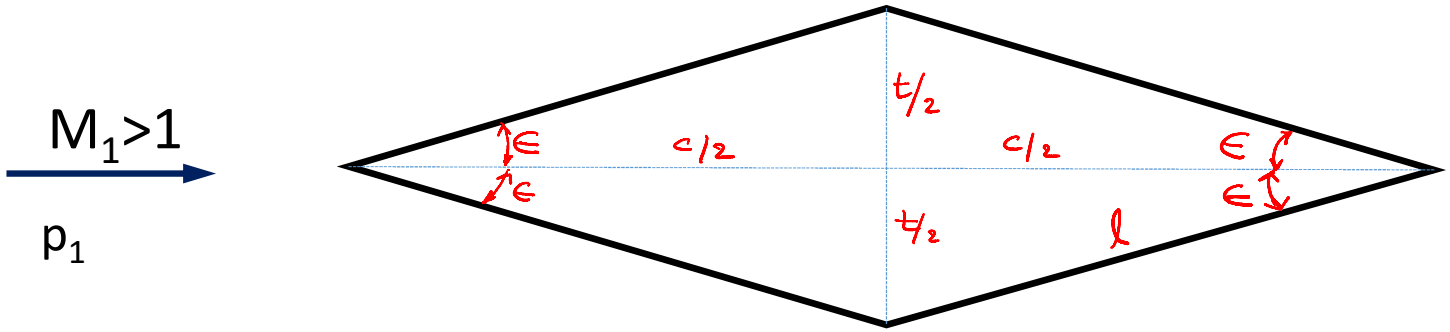
- Whenever we have a body made up of straight-line segments and the deflection angles are small enough so that no detached shock waves occur, the flow over the body goes through a series of distinct oblique shock and expansion waves.

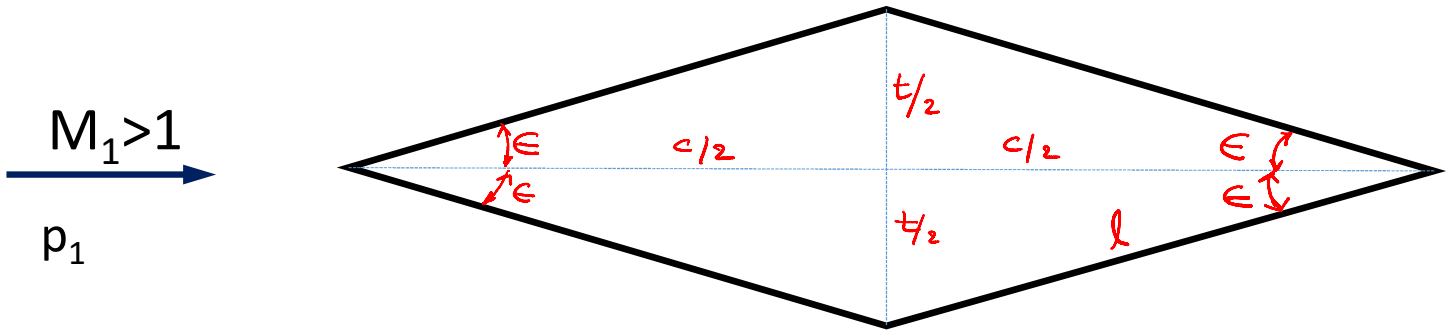
- The pressure distribution on the surface (hence the lift and drag) can be obtained *exactly* from both the shock- and expansion wave theories.

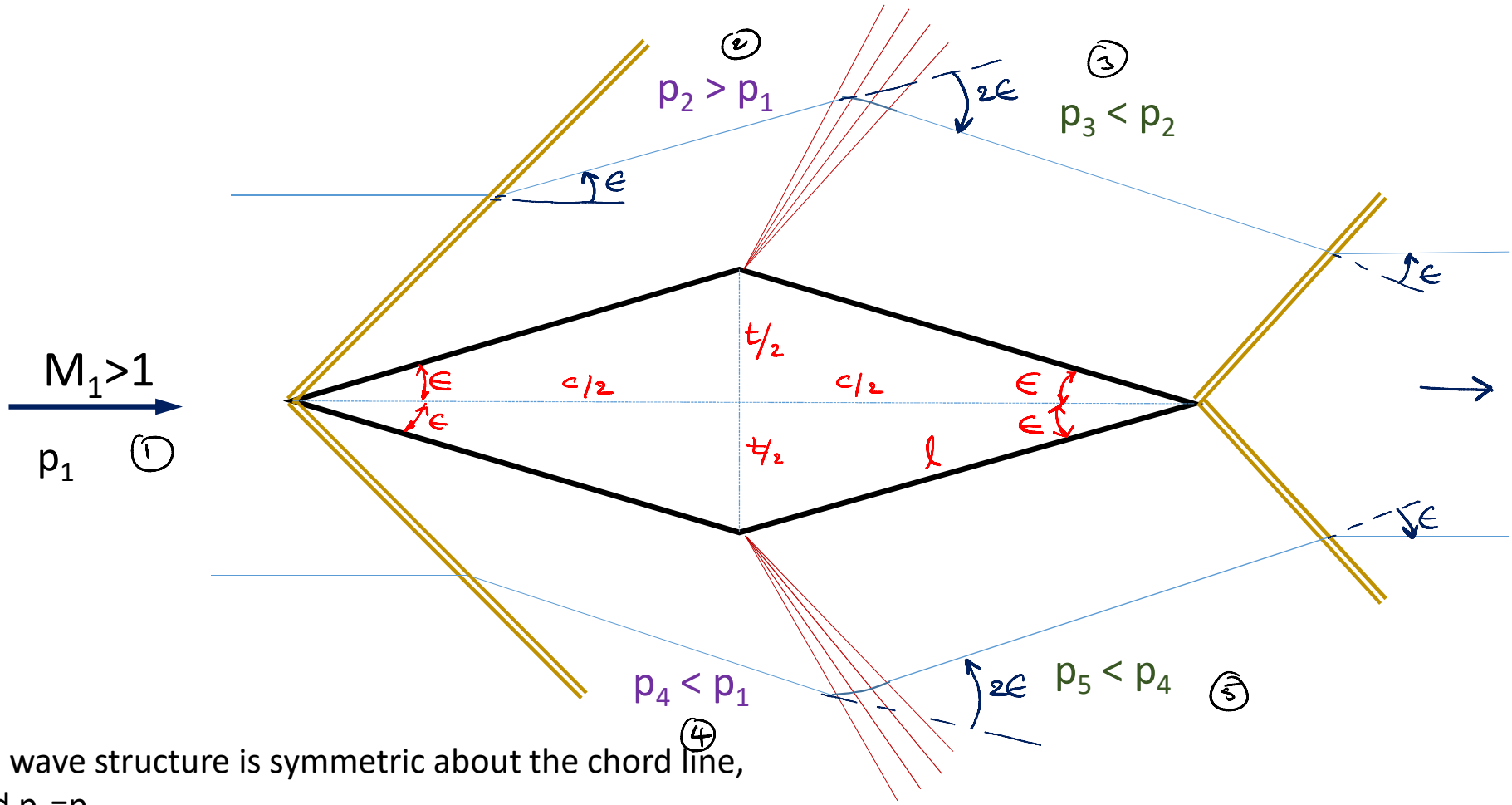
- The inviscid, supersonic flow over a flat plate at angle of attack is exactly given by the combination of shock and expansion waves

Diamond airfoil

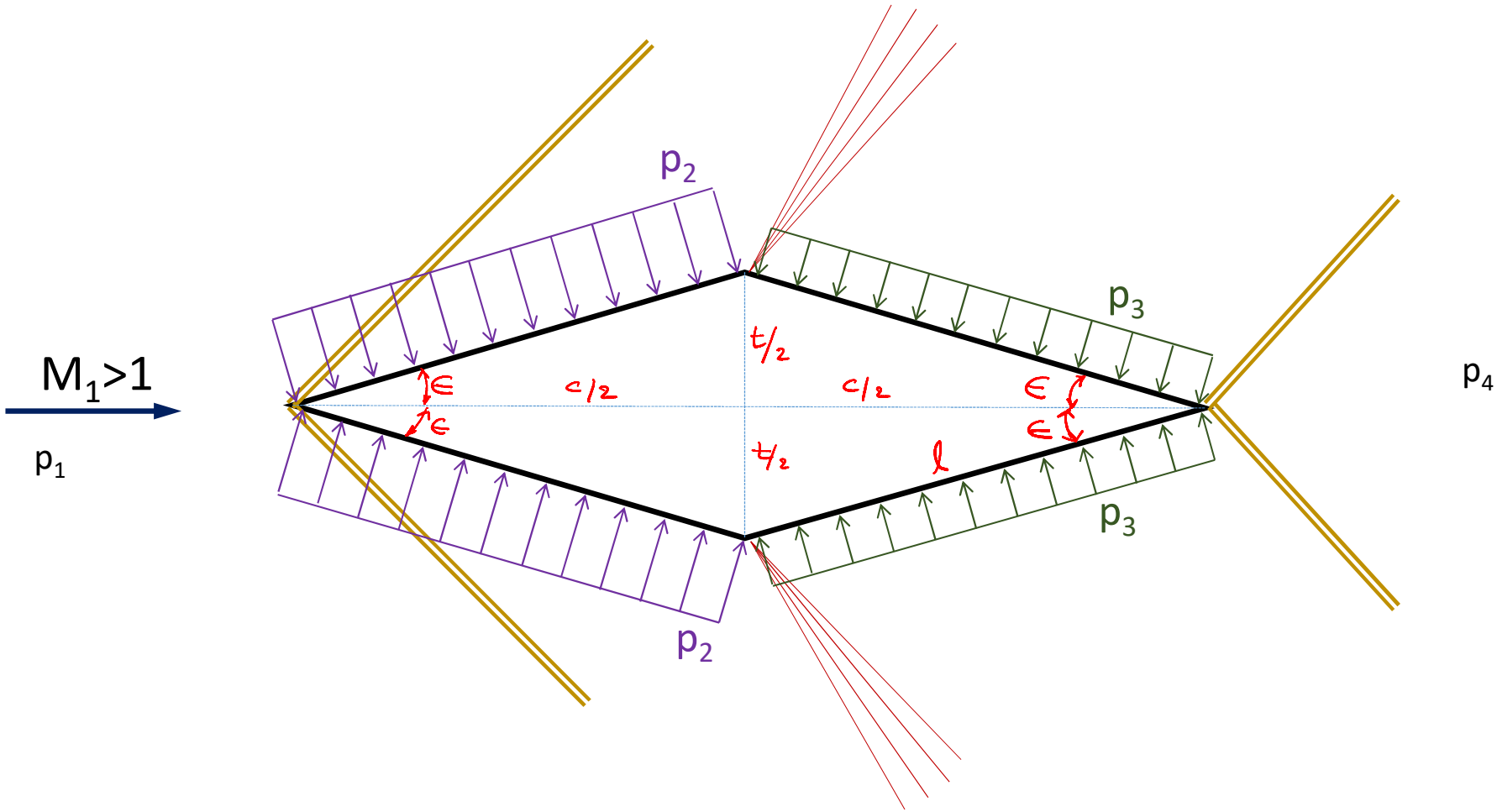
- Diamond airfoil:
  - Chord c
  - Maximum thickness t
  - Side length l
  - Wedge half-angle ε
  - *Zero* angle of attack
- Flow is symmetric

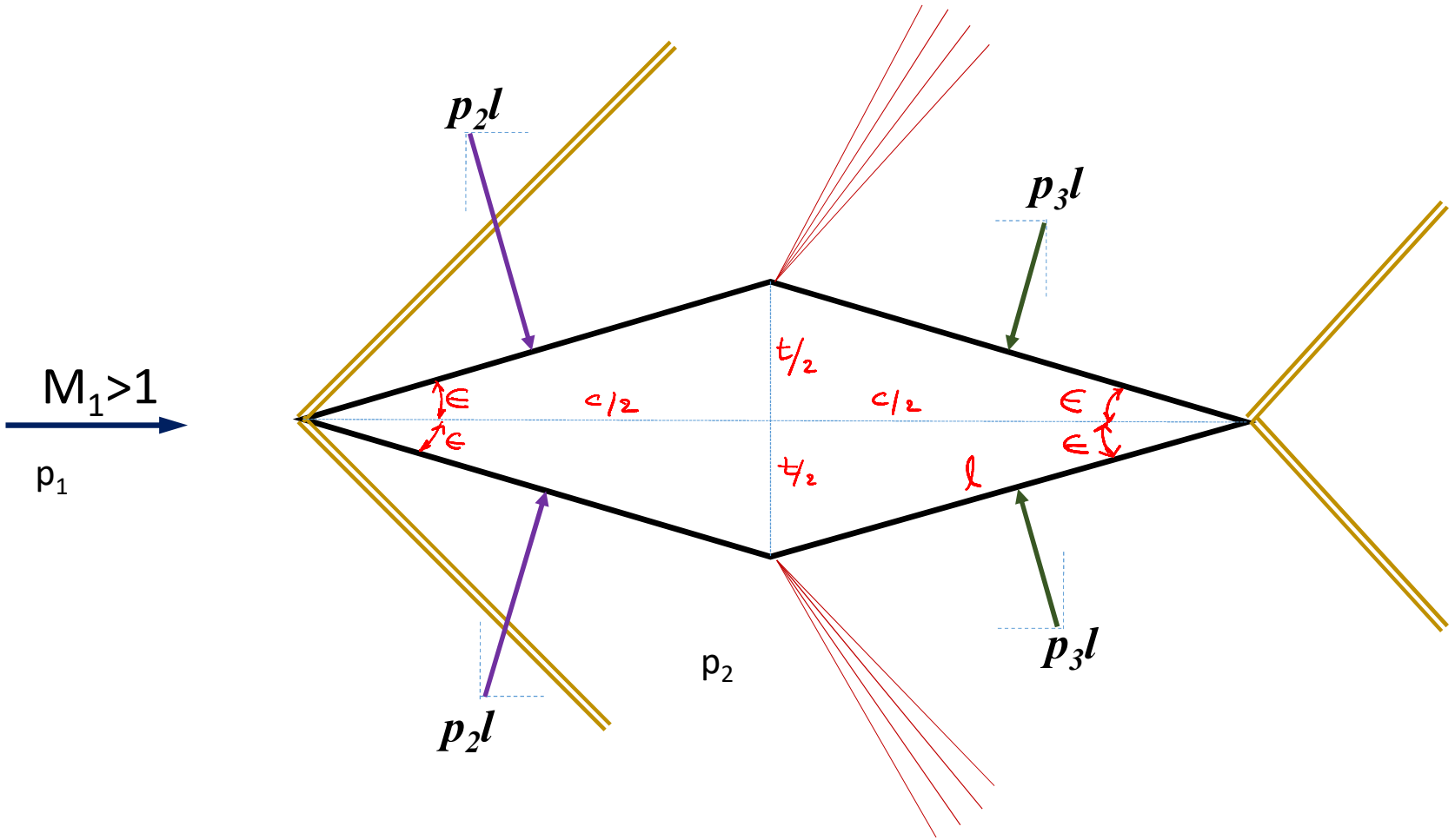



M1>1
p1
t/2
c/2
c/2
ϵ
ϵ
ϵ
ϵ
t/2
l



Since the wave structure is symmetric about the chord line, $p_2=p_4$ and $p_3=p_5$


M1>1
p1
p2
p3
p4
t/2
c/2
c/2
ϵ
ϵ
ϵ
ϵ
t/2
l
p3
p2


M1>1
p1
p2l
p3l
t/2
c/2
c/2
ϵ
ϵ
ϵ
ϵ
t/2
l
p2
p3l
p2l

LIFT:

$$L = -p_2 l\cos\epsilon + p_2 l\cos\epsilon - p_3 l\cos\epsilon + p_3 l\cos\epsilon$$

$$\Rightarrow \quad L = 0$$

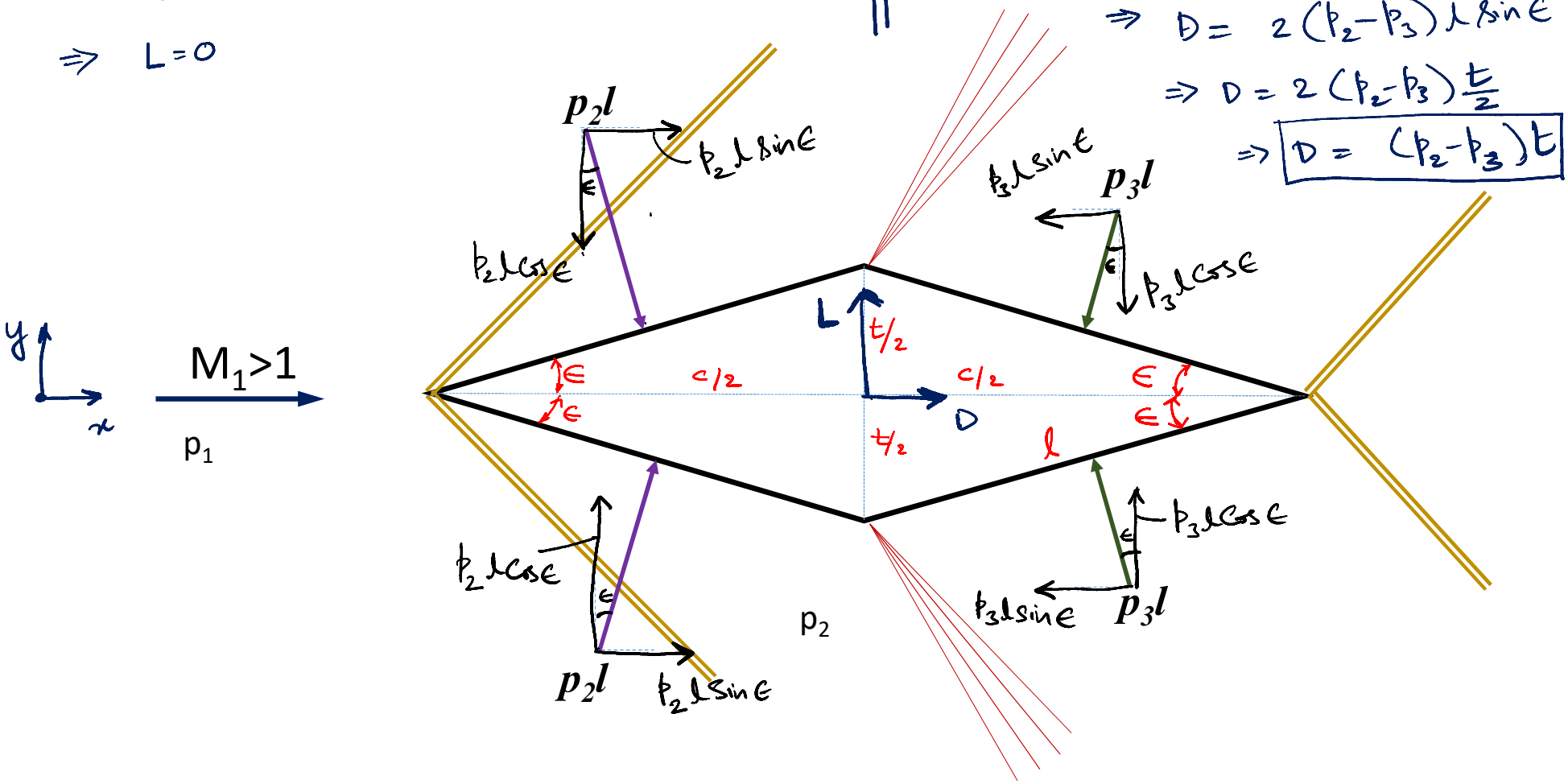
DRAG:

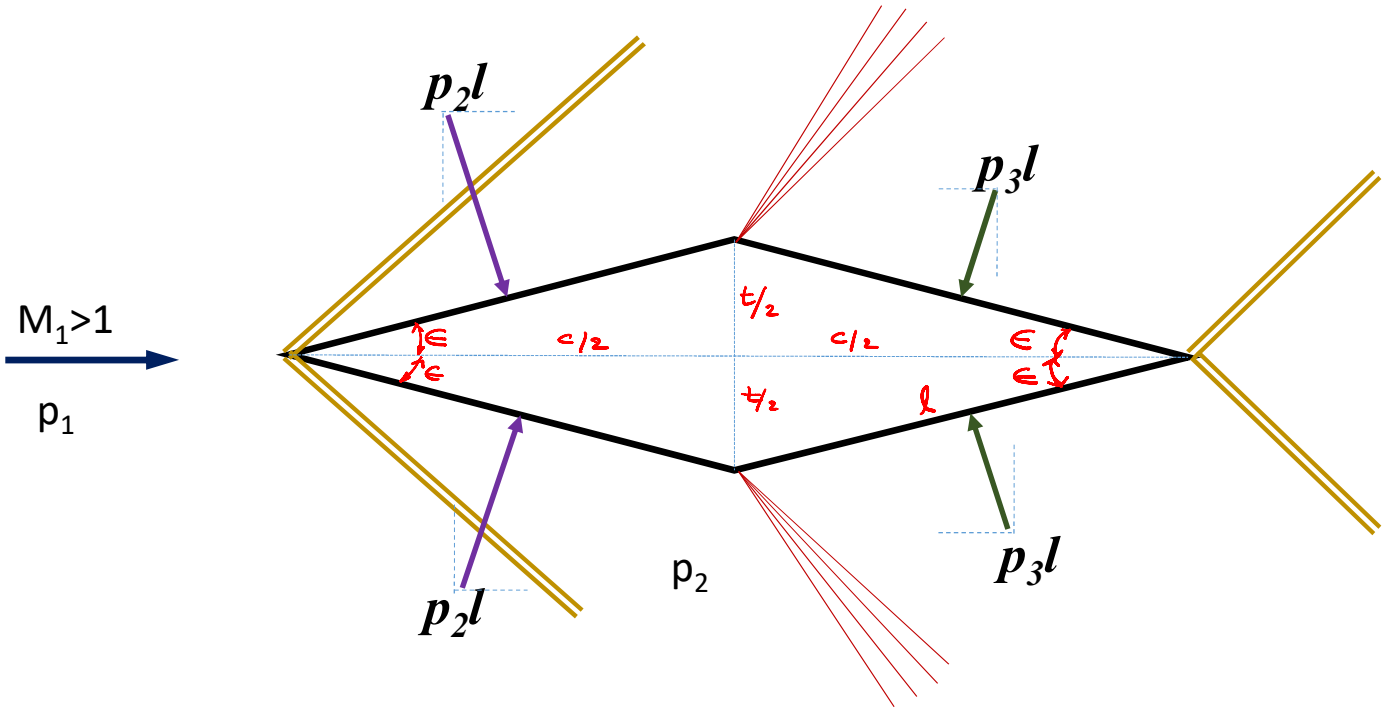
$$D = p_2 l\sin\epsilon - p_3 l\sin\epsilon + p_2 l\sin\epsilon - p_3 l\sin\epsilon$$

$$\Rightarrow \quad D = 2(p_2 - p_3) l\sin\epsilon$$

$$\Rightarrow \quad D = 2(p_2 - p_3)\frac{t}{2}$$

$$\Rightarrow \quad \boxed{D = (p_2 - p_3)t}$$





$$L' = 0$$

$$D' = 2(p_2 l \sin\varepsilon - p_3 l \sin\varepsilon) = 2(p_2 - p_3)\frac{t}{2}$$

$$D' = (p_2 - p_3)t$$

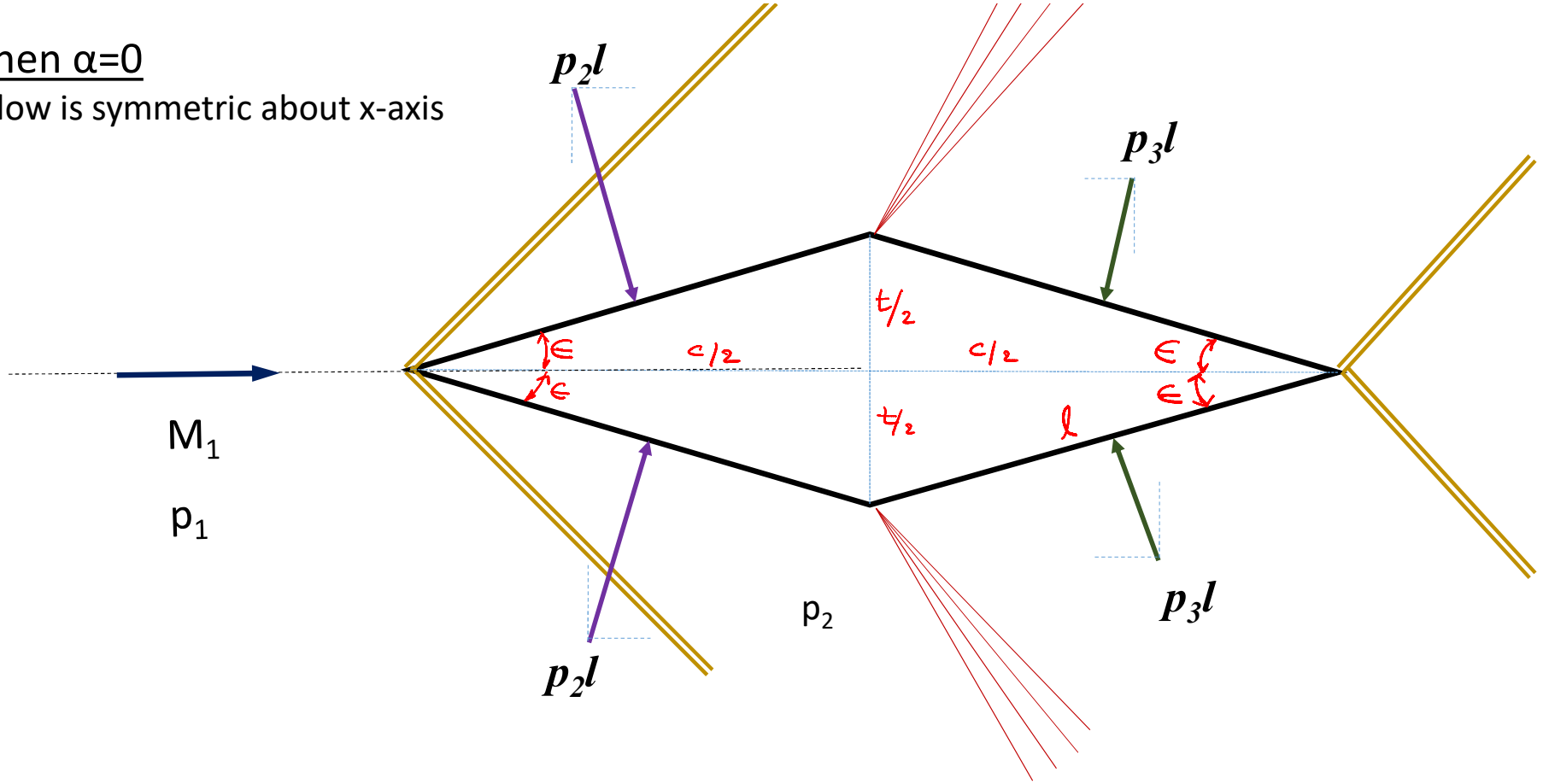
A note on Drag in Inviscid, Supersonic flow over 2D bodies….

- In an inviscid, incompressible flow over 2D bodies, Drag per unit span is predicted to be zero from potential theory - *d'Alembert's paradox*
- But, in an inviscid, supersonic flow over 2D bodies, Drag per unit span is finite as observed here
- This new source of drag is called *wave drag* and it arises purely because of shock waves
- The existence of wave drag is inherently related to the increase in entropy and consequently to the loss of total pressure across the oblique shock waves created by the airfoil.

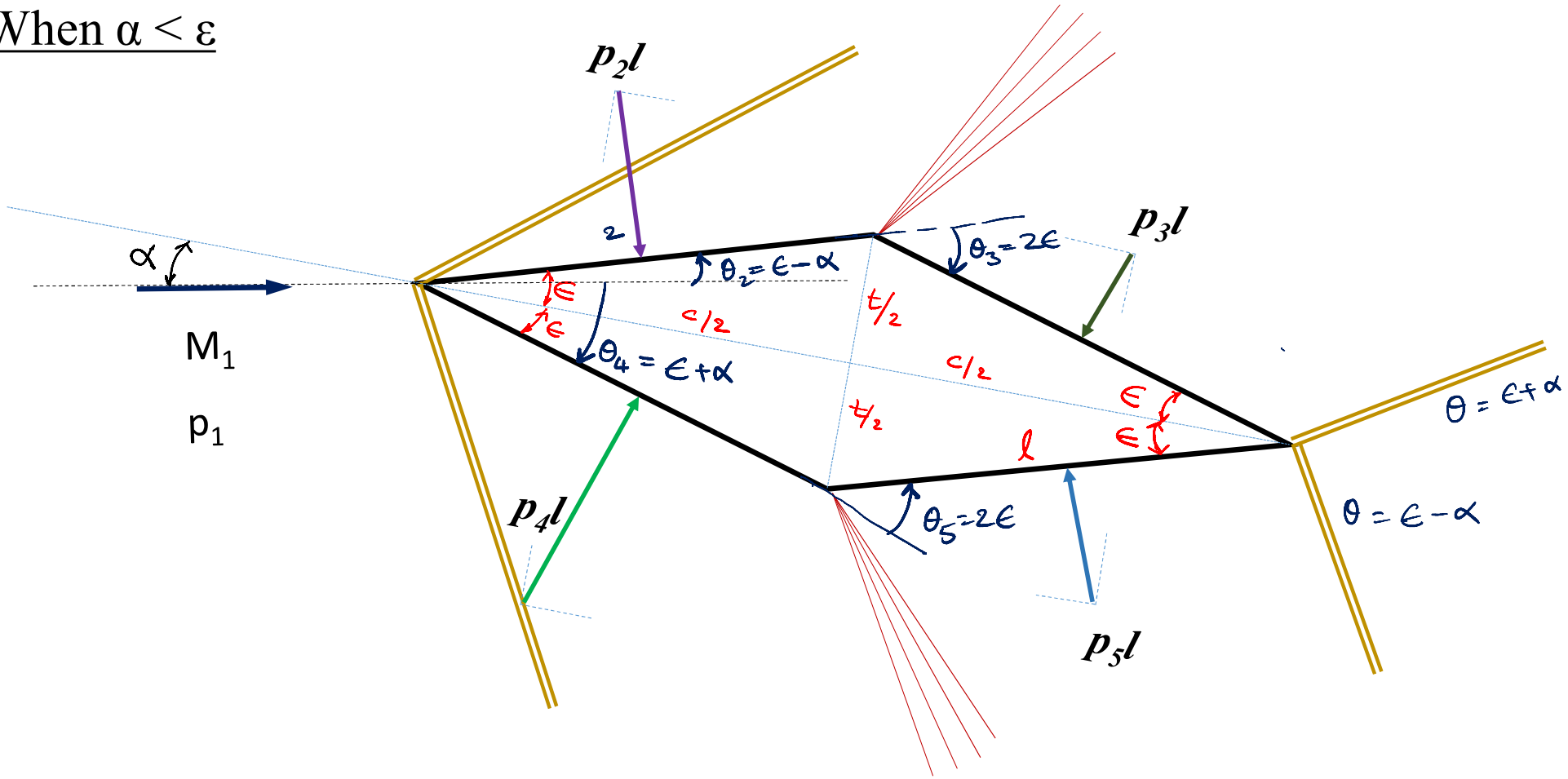
Flow over a diamond airfoil with increase in angle of attack

When α=0

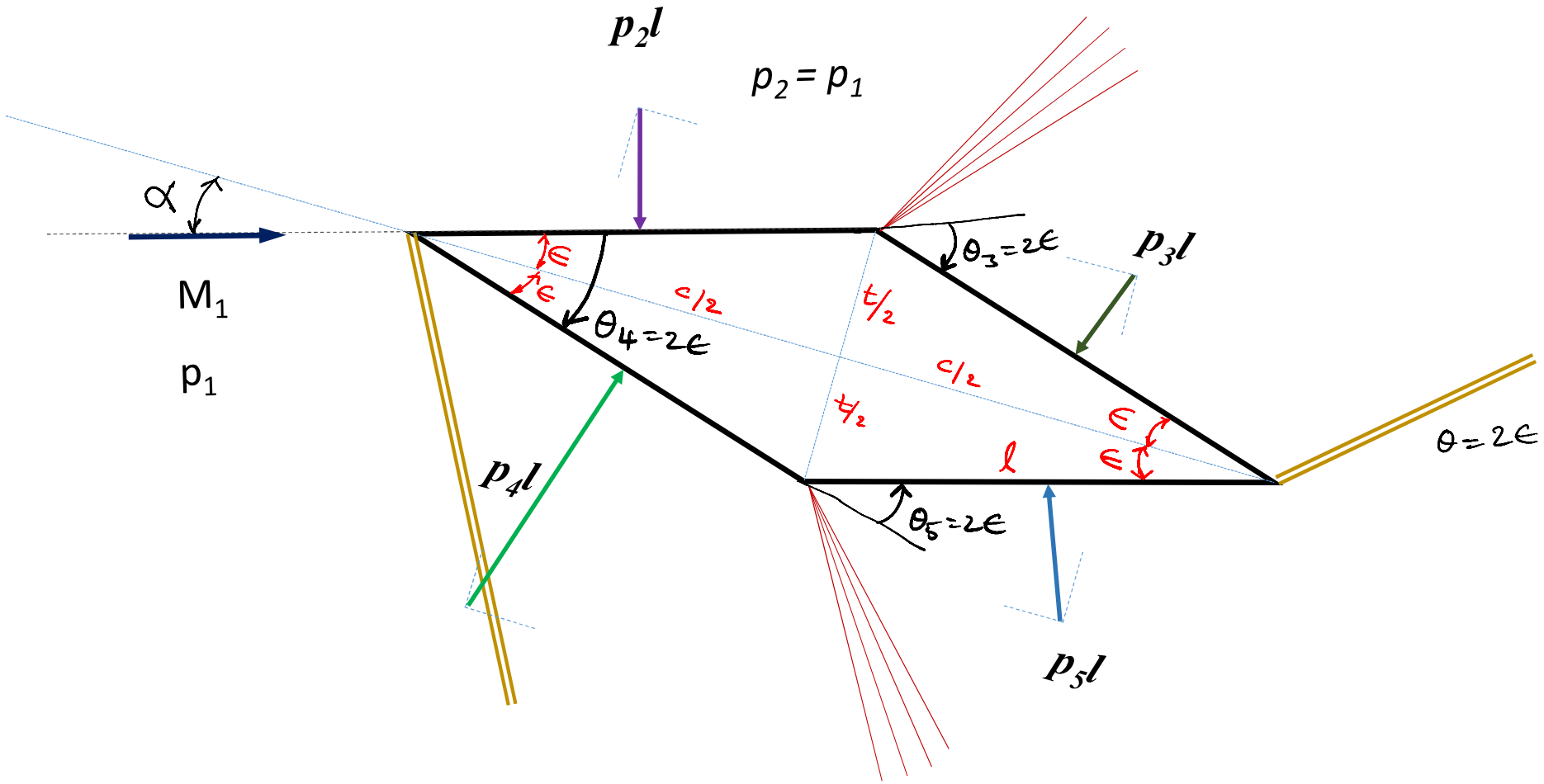
– Flow is symmetric about x-axis



When $\alpha < \varepsilon$



When $\alpha = \varepsilon$



When $\alpha > \varepsilon$

